

5_3_Logowanie_danych_i_obsługa_narzędzi_graficznych_zadania-v2

April 6, 2022

Do wykonania ćwiczeń konieczne jest uruchomienie symulacji z turtlesim dla instrukcji 5.

1 Rejestrowanie rosbaga

Należy wykorzystać polecenia z terminala.

1.0.1 Zadanie 1

Zarejestrować rosbaga dla topiców od położenia robotów. Nagrywanie przerwać po 10s.

```
[1]: !roslab record /robot1/pose /robot2/pose /robot3/pose --duration=10
```

```
[ INFO] [1649224214.227673314]: Subscribing to /robot1/pose
[ INFO] [1649224214.232409473]: Subscribing to /robot2/pose
[ INFO] [1649224214.235860089]: Subscribing to /robot3/pose
[ INFO] [1649224214.239289612]: Recording to '2022-04-06-05-50-14.bag'.
```

```
[ ]: !roslab record --duration=10 /robot1/pose /robot2/pose /robot3/pose
```

```
[2]: !roslab info '2022-04-06-05-50-14.bag'
```

```
path:      2022-04-06-05-50-14.bag
version:    2.0
duration:   9.8s
start:      Apr 06 2022 05:50:14.44 (1649224214.44)
end:        Apr 06 2022 05:50:24.23 (1649224224.23)
size:       146.1 KB
messages:   1839
compression: none [1/1 chunks]
types:      turtlesim/Pose [863b248d5016ca62ea2e895ae5265cf9]
topics:     /robot1/pose    613 msgs    : turtlesim/Pose
            /robot2/pose    613 msgs    : turtlesim/Pose
            /robot3/pose    613 msgs    : turtlesim/Pose
```

1.0.2 Zadanie 2

Zarejestrować rosbaga o nazwie robot_poses.bag rejestrującego położenie robotów. Zakończenie rejestrowania odbywa się ręcznie przez użytkownika przy pomocy ctrl+c.

```
[3]: !rosbag record /robot1/pose /robot2/pose /robot3/pose -o robot_poses
```

```
[ INFO] [1649224453.283914563]: Subscribing to /robot1/pose
[ INFO] [1649224453.288638369]: Subscribing to /robot2/pose
[ INFO] [1649224453.292592539]: Subscribing to /robot3/pose
[ INFO] [1649224453.296158323]: Recording to 'robot_poses.bag'.
^C
```

1.0.3 Zadanie 3

Zarejestrować rosbaga z prędkościami robotów trwającego 5s. Należy zapisywać tylko 3 ostatnie rosbagi. Pozostałe mają się kasować.

```
[4]: !rosbag record /robot1/cmd_vel /robot2/cmd_vel /robot3/cmd_vel --duration=5s
↪--split --max-splits=3
```

```
[ INFO] [1649224594.748296491]: Subscribing to /robot1/cmd_vel
[ INFO] [1649224594.753067390]: Subscribing to /robot2/cmd_vel
[ INFO] [1649224594.756133535]: Subscribing to /robot3/cmd_vel
[ INFO] [1649224594.759669366]: Recording to
'2022-04-06-05-56-34_0.bag'.
[ INFO] [1649224599.762110248]: Closing '2022-04-06-05-56-34_0.bag'.
[ INFO] [1649224599.770527510]: Recording to
'2022-04-06-05-56-39_1.bag'.
[ INFO] [1649224604.773046815]: Closing '2022-04-06-05-56-39_1.bag'.
[ INFO] [1649224604.781933062]: Recording to
'2022-04-06-05-56-44_2.bag'.
[ INFO] [1649224609.784696570]: Closing '2022-04-06-05-56-44_2.bag'.
[ INFO] [1649224609.791638324]: Recording to
'2022-04-06-05-56-49_3.bag'.
[ INFO] [1649224614.794036823]: Closing '2022-04-06-05-56-49_3.bag'.
[ INFO] [1649224614.802878772]: Recording to
'2022-04-06-05-56-54_4.bag'.
^C
```

1.0.4 Zadanie 4

Zarejestrować dla każdego topic'a od prędkości robotów tylko 20 wiadomości.

```
[5]: !rosbag record /robot1/cmd_vel /robot2/cmd_vel /robot3/cmd_vel -l 20
```

```
[ INFO] [1649224683.852131584]: Subscribing to /robot1/cmd_vel
[ INFO] [1649224683.856266270]: Subscribing to /robot2/cmd_vel
[ INFO] [1649224683.859292526]: Subscribing to /robot3/cmd_vel
[ INFO] [1649224683.862683452]: Recording to '2022-04-06-05-58-03.bag'.
```

```
[6]: !rosbag info '2022-04-06-05-58-03.bag'
```

```
path:          2022-04-06-05-58-03.bag
version:       2.0
duration:      0.3s
start:         Apr 06 2022 05:58:27.99 (1649224707.99)
end:          Apr 06 2022 05:58:28.25 (1649224708.25)
size:          15.5 KB
messages:      60
compression:   none [1/1 chunks]
types:         geometry_msgs/Twist [9f195f881246fdfa2798d1d3eebca84a]
topics:        /robot1/cmd_vel      20 msgs      : geometry_msgs/Twist
               /robot2/cmd_vel      20 msgs      : geometry_msgs/Twist
               /robot3/cmd_vel      20 msgs      : geometry_msgs/Twist
```

1.0.5 Zadanie 5

Zarejestrować wszystkie topic'i dla node'a turtlesim.

```
[7]: !rosbag record --node=turtlesim
```

```
Recording from: turtlesim
[ INFO] [1649224818.792627115]: Recording to '2022-04-06-06-00-18.bag'.
[ INFO] [1649224818.825981731]: Subscribing to /robot1/cmd_vel
[ INFO] [1649224818.830508832]: Subscribing to /robot2/cmd_vel
[ INFO] [1649224818.834531889]: Subscribing to /robot3/cmd_vel
^C
```

2 Odtwarzanie rosbaga

2.0.1 Zadanie 6

Odtworzyć z rosbaga TSR.bag dane dla topic'u /robot1/cmd_vel.

```
[8]: !rosbag info 'TSR.bag'
```

```
path:          TSR.bag
version:       2.0
duration:      9.9s
start:         Apr 06 2022 05:40:45.36 (1649223645.36)
end:          Apr 06 2022 05:40:55.30 (1649223655.30)
```

```

size:          2.4 MB
messages:      25222
compression:   none [3/3 chunks]
types:         geometry_msgs/Twist [9f195f881246fdfa2798d1d3eebca84a]
               rosgraph_msgs/Log   [acffd30cd6b6de30f120938c17c593fb]
               turtlesim/Color     [353891e354491c51aabe32df673fb446]
               turtlesim/Pose       [863b248d5016ca62ea2e895ae5265cf9]
topics:        /robot1/cmd_vel      7157 msgs    : geometry_msgs/Twist
               /robot1/color_sensor 610 msgs     : turtlesim/Color
               /robot1/pose         610 msgs     : turtlesim/Pose
               /robot2/cmd_vel      7228 msgs    : geometry_msgs/Twist
               /robot2/color_sensor 610 msgs     : turtlesim/Color
               /robot2/pose         610 msgs     : turtlesim/Pose
               /robot3/cmd_vel      7168 msgs    : geometry_msgs/Twist
               /robot3/color_sensor 610 msgs     : turtlesim/Color
               /robot3/pose         610 msgs     : turtlesim/Pose
               /rosout              9 msgs        : rosgraph_msgs/Log

```

```
[10]: !rosbag play "TSR.bag" --topics /robot1/cmd_vel
```

```
[ INFO] [1649225013.820460905]: Opening TSR.bag
```

Waiting 0.2 seconds after advertising topics... done.

Hit space to toggle paused, or 's' to step.

```
[RUNNING] Bag Time: 1649223655.295134 Duration: 9.673423 / 9.677111
Done.
```

2.0.2 Zadanie 7

Odtworzyć rosbaga z dwukrotnie większą prędkością.

```
[19]: !rosbag play --hz=200 "TSR.bag" --topics /robot1/cmd_vel
```

```
[ INFO] [1649225247.300433992]: Opening TSR.bag
```

Waiting 0.2 seconds after advertising topics... done.

Hit space to toggle paused, or 's' to step.

```
[RUNNING] Bag Time: 1649223655.295167 Duration: 9.673457 / 9.677111
Done.
```

2.0.3 Zadanie 8

Rozpocząć odtwarzanie rosbaga od 5 sekundy.

```
[21]: !rosbag play "TSR.bag" --topics /robot1/cmd_vel -s 5
```

[INFO] [1649225344.344414704]: Opening TSR.bag

Waiting 0.2 seconds after advertising topics... done.

Hit space to toggle paused, or 's' to step.

[RUNNING] Bag Time: 1649223655.295147 Duration: 4.935364 / 4.939038
Done.

[]: